P04: Makers Makin' It, Act II – The Seequel
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PROJECT NAME: "FAANG+ Stock Explorer"
DESCRIPTION: The website will allow users to interactively explore and compare the major tech stocks, and market leaders AAPL, MSFT, AMZN, GOOGL, META, and NVDA. We will present stock data through dynamic visualizations and analyze what differentiates these stocks in terms of growth over time, risk, and market behavior. Users will also be able to compare these influential stocks with other non-FAANG+ stocks.

TARGET SHIP DATE: 2026-04-20

Program Components

Flask App

- ☐ init.py (main app + routes)
  - ☐ creates Flask app, config, and session handling
  - ☐ renders templates and connects frontend to backend logic
- ☐ /register
  - ☐ adds a new user to the users table
  - ☐ checks if username is unique & hashes password
  - ☐ stores session, redirects to *dashboard*
- ☐ /login
  - ☐ verifies user credentials
  - ☐ stores session, redirects to *dashboard*
- ☐ /logout
  - ☐ clears session, redirects to *login*
- ☐ /dashboard
  - ☐ main analytics page
  - ☐ loads selected stocks and displays visualizations of metrics, e.g. indexed growth and volume over time
- ☐ /explore
  - ☐ interactive exploration page where users select and compare FAANG+ and non-FAANG+ stocks.
  - ☐ Upon entry/selection, redirects to *stock_viewer*

- ☐ /supply_chain
  - ☐ Provides visualization of correlation between three stocks in a small case study: TSM, NVDA, and TSLA
- ☐ /analysis
  - ☐ Displays correlation matrix of the prices over time of our six FAANG+ stocks.
  - ☐ Displays visualization of a bar graph measuring risk-adjusted returns.
- ☐ /stock_viewer
  - ☐ displays detailed profile for a single stock or a comparison between two selected stocks

Database

- ☐ data.py connects to sqlite3 DB to create & maintain tables:
  - ☐ users: user account info
  - ☐ stocks: stock metadata (tickers, names), FAANG+ & user-added
  - ☐ stock_data: cached historical data (from dataset + API)

Templates

- ☐ login.html username + password form
- ☐ register.html account creation form
- ☐ dashboard.html
  - ☐ Indexed growth chart
  - ☐ Volume over time chart
- ☐ explore.html lets user select a stock or two to compare
- ☐ stock_viewer.html individual/comparison stock profile
- ☐ analysis.html analysis of specific FAANG+ stocks
  - ☐ Correlation matrix
  - ☐ Risk-adjusted returns bar chart
- ☐ supply_chain.html case study of three related stocks

Database Organization

USERS			
INTEGER	id	PK	Unique id
TEXT	username		
TEXT	password		

STOCKS			
INTEGER	id	PK	Unique, Auto-increment
TEXT	ticker		Few letters representing a stock
TEXT	company_name		

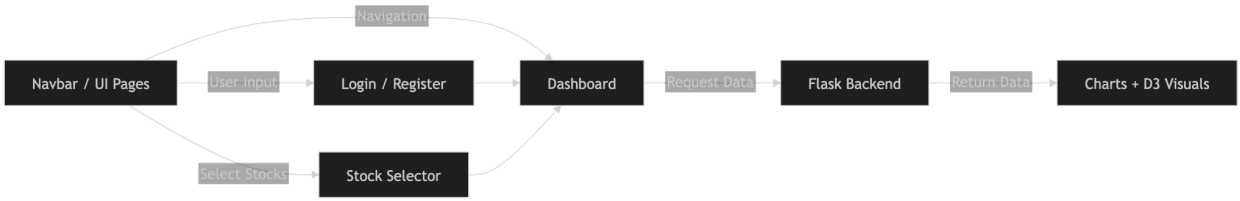
STOCK_DATA			
INTEGER	stock_id	FK	
TIMESTAMP	date		
FLOAT	open_price		Stock price upon market open
FLOAT	close_price		Stock price upon market close
FLOAT	returns		Gains/losses between consecutive days
INTEGER	volume		Number of stock multiplied by price of stock

Frontend Framework

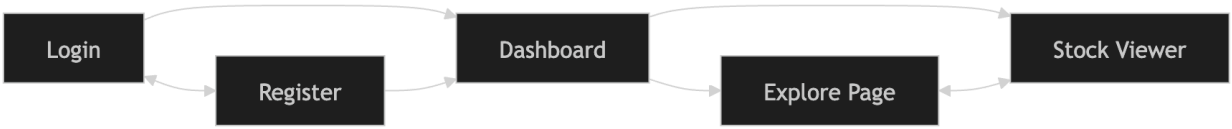
- ☐ **Tailwind** (styling because our devos have more experience with it)
- ☐ **JavaScript**
  - ☐ dynamically updates charts based on selected stocks
  - ☐ handles user interactions (dropdowns, toggles, comparisons)
  - ☐ sends requests to Flask backend for updated data
  - ☐ integrates Chart.js and D3.js visualizations

Component Map

- ☐ **Navbar** navigation across pages
- ☐ **Login & Register**
- ☐ **Dashboard Panel** main visualizations
- ☐ **Chart Components** Chart.js graphs
- ☐ **D3 Component** animated growth race
- ☐ **Tailwind**
- ☐ **Stock Selector** dropdown/multi-select with FAANG+ and external stocks



Site Map



Data Flow

Task Breakdown

- ☐ Flask App - Robert Chen
    - ☐ Login & Register
    - ☐ Dashboard
    - ☐ Explore Page
    - ☐ Stock Viewer
- ☐ Visualizations - Kalimul Kaif
    - ☐ Chart.js
    - ☐ D3.js
- ☐ Database - Cody Wong
    - ☐ Dataset Handling
    - ☐ DB Organization
    - ☐ Integrations with site features
- ☐ Stock Analysis Features - Zixi Qiao
    - ☐ Stock-to-Stock Comparisons
    - ☐ Metrics over time

REST API SECTION

- ☐ https://developer.yahoo.com/api/

Dataset Handling from Backend to Frontend

- In static folder as csv
- In build_db.py, parse features into STOCK_DATA
- Extract into __init__.py for dashboard and analysis routes
- Use Pandas to convert the raw data into lists of dictionaries for our two charts (correlation matrix and Sharpe ratio bar graph)
- Pass into javascript using jsonify to build the charts